

# Quick Check

## Lesson 4: Meiosis and Its Stages: Why Do Sperm and Egg Cells Have 23 Chromosomes?

*7 multiple-choice questions • choose A, B, C or D*

Sub-Strand 1.4: Cell Division | Lesson 4

**A human body cell has 46 chromosomes. Suppose a sperm cell and an egg cell each also had 46 chromosomes. How many chromosomes would the cell formed at fertilisation (the zygote) have?**

**A** 46

**B** 92

**C** 138

**D** 23

**During meiosis I, which event happens in Anaphase I?**

- A** Chromosomes copy their DNA again
- B** The cell membrane splits to make four cells at once
- C** Sister chromatids separate and move to opposite poles
- D** Homologous chromosomes separate and move to opposite poles

## Which statement correctly describes Anaphase II of meiosis?

- A** Chromosomes line up as pairs at the middle of the cell
- B** Sister chromatids separate and move to opposite poles
- C** Homologous chromosomes separate for the first time
- D** Homologous chromosomes pair up and cross over

**A human body cell with 46 chromosomes starts meiosis. What is found in each cell at the end of meiosis I (after Telophase I)?**

- A** 46 chromosomes, each with one chromatid
- B** 46 chromosomes, each with two chromatids
- C** 23 chromosomes, each with two chromatids
- D** 23 chromosomes, each with one chromatid

## Why do sperm and egg cells each have 23 chromosomes instead of 46?

- A** Meiosis halves the chromosome number, and fertilisation restores 46 in the zygote
- B** Mitosis makes them, so they have the same number as body cells
- C** They have 23 because the zygote needs only 23 chromosomes to grow
- D** They have 46 chromosomes, and the extra 23 are thrown away after fertilisation

**Which row correctly compares meiosis in the reproductive organs with mitosis in the muscle cells of a Kenyan runner?**

- A** Meiosis: one division, two identical cells with 46. Mitosis: two divisions, four different cells with 23
- B** Meiosis and mitosis both give two identical cells with 46 chromosomes
- C** Meiosis: two divisions, four genetically different cells with 23. Mitosis: one division, two identical cells with 46
- D** Meiosis: two divisions, two identical cells with 46. Mitosis: one division, four cells with 23

**In a model of the human life cycle, a zygote with 46 chromosomes grows into a teenager. Which process produces the billions of new body cells for this growth, and how many chromosomes does each new body cell have?**

- A** Mitosis, giving cells with 46 chromosomes
- B** Fertilisation, giving cells with 92 chromosomes
- C** Meiosis, giving cells with 23 chromosomes
- D** Meiosis, giving cells with 46 chromosomes

# All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).